

Outline

Motivation: The Integrals We Cannot Solve

The Monte Carlo Principle

From Integral to Expectation

Crude Monte Carlo: Algorithm

Worked Toy Examples

Law of Large Numbers and Monte Carlo Error

Crude Monte Carlo in Practice

Importance Sampling

Variance Reduction Ideas

Practical Monte Carlo Diagnostics

Unit Summary

$$\Rightarrow \theta = \int_{\mathcal{X}} h(x) dx \quad \mathcal{X} = \mathbb{R}, \mathbb{R}^2, \mathbb{R}^d$$

$$\Rightarrow \theta = \int \frac{h(x)}{f(x)} \cdot f(x) dx \quad f: \text{density}$$

$$= \mathbb{E}_f \left[\frac{h(x)}{f(x)} \right] \rightarrow \text{average}$$

$$\Rightarrow X_1 \dots X_n \sim f$$

$$\Rightarrow \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n \frac{h(X_i)}{f(X_i)}$$

?? how good is this estimator $\hat{\theta}_n$??

$\Rightarrow$ 1) unbiasedness

2) consistency

3) efficiency

4) WLLN / SLLN / CLT

$$\Rightarrow 1) \mathbb{E}[\hat{\theta}_n] = \theta \rightarrow \text{Recall the proof}$$

$$\stackrel{\text{Ex.}}{\Leftrightarrow} \theta = \int_0^1 \frac{\sin x}{x} dx = \int_0^1 \frac{\sin x}{\frac{1}{x}} dx = \mathbb{E}_f \left[\frac{\sin x}{x} \right]$$

$X \sim f = U[0,1]$

$$\Rightarrow \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n \frac{\sin x_i}{x_i} \rightarrow \text{estimator of } \theta.$$

Rewriting an Integral as an Expectation

Any integral of the form

$$\theta = \int_{\mathcal{X}} g(x) dx$$

can be rewritten by introducing *any* density f with $f(x) > 0$ wherever $g(x) \neq 0$:

$$\theta = \int_{\mathcal{X}} \frac{g(x)}{f(x)} f(x) dx = E_f \left[\frac{g(X)}{f(X)} \right], \quad X \sim f.$$

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Special case: f already a density

If $\theta = \int h(x)f(x) dx$ with f a genuine probability density, simply take $g(x)/f(x) = h(x)$:

$$\theta = E_f[h(X)].$$

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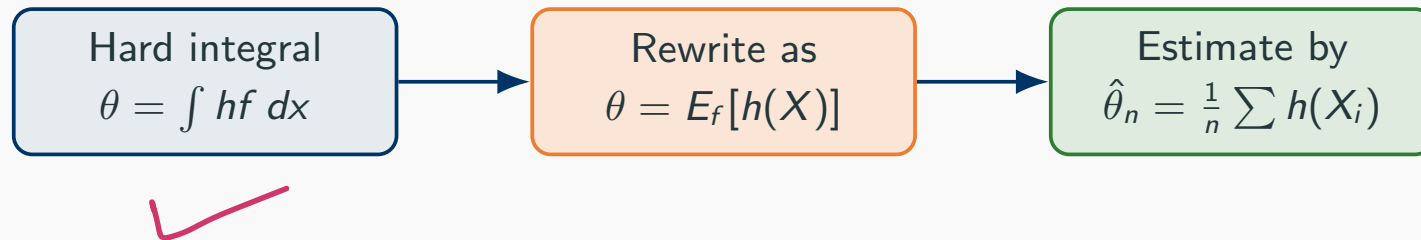
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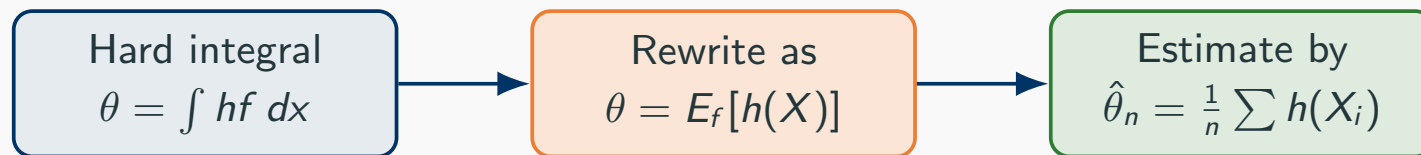
$$\theta = E_f[h(X)].$$

This is the **entire trick** of Monte Carlo: turn “integrate” into “take an expectation”, then estimate the expectation with a sample average.

The Monte Carlo Principle: Integral $\rightarrow$ Expectation $\rightarrow$ Average

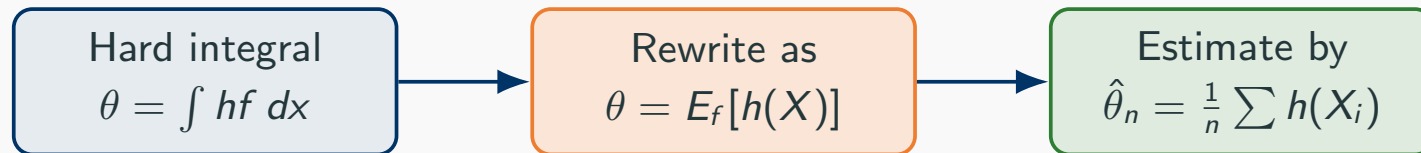


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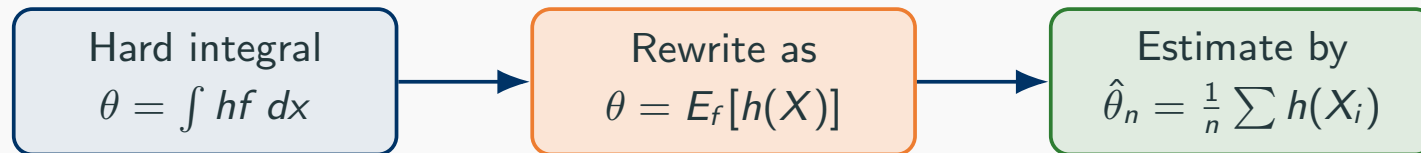
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- Step 1 (integral $\rightarrow$ expectation) is pure algebra – always possible.
- Step 2 (expectation $\rightarrow$ average) requires **Module 1**: we must be able to draw $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$.
- Everything in Module 2 is about making Step 2 as accurate and efficient as possible.

Crude (Basic) Monte Carlo Estimator

Algorithm: Crude Monte Carlo

1. Generate $X_1, X_2, \dots, X_n \stackrel{\text{iid}}{\sim} f$.
2. Compute $h(X_1), \dots, h(X_n)$.
3. Return

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i).$$

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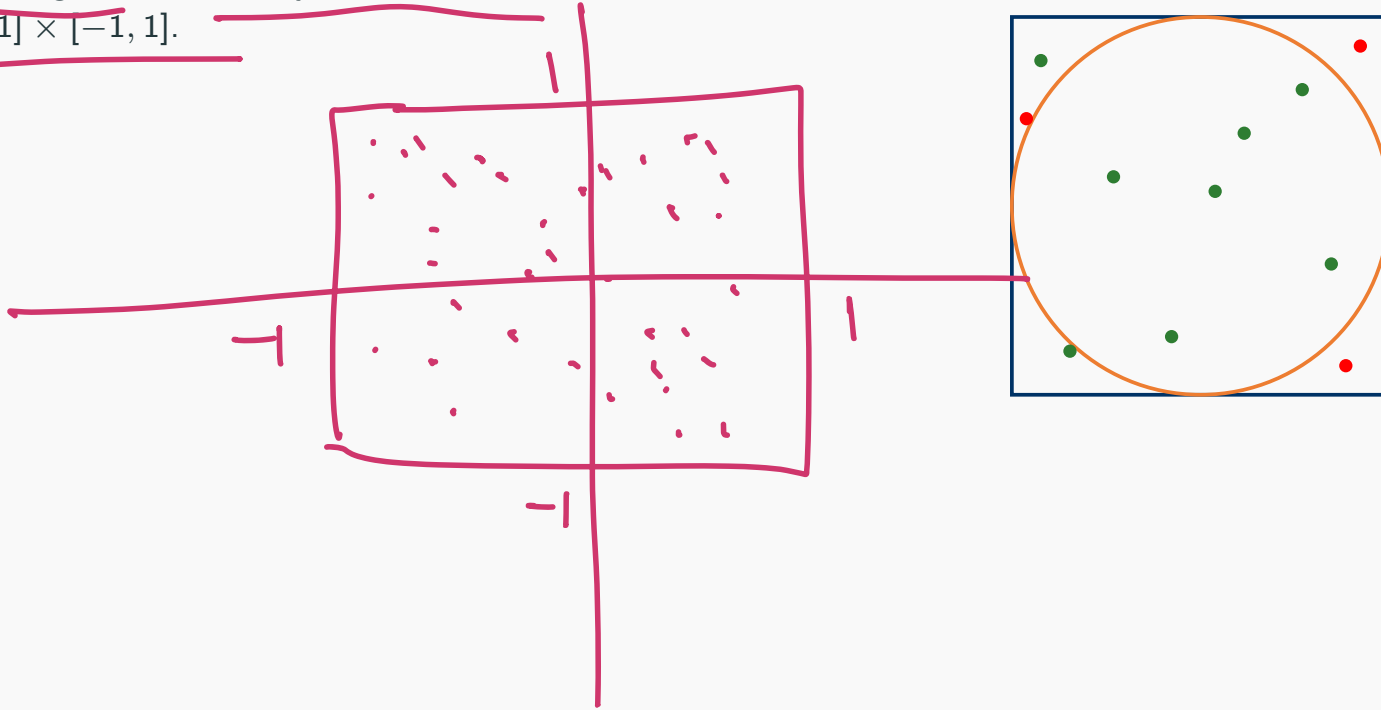
Unbiasedness


$$E[\hat{\theta}_n] = \frac{1}{n} \sum_{i=1}^n E[h(X_i)] = \frac{1}{n} \cdot n \cdot E_f[h(X)] = \theta.$$

So $\hat{\theta}_n$ is an **unbiased** estimator of θ , for *any* sample size $n \geq 1$.

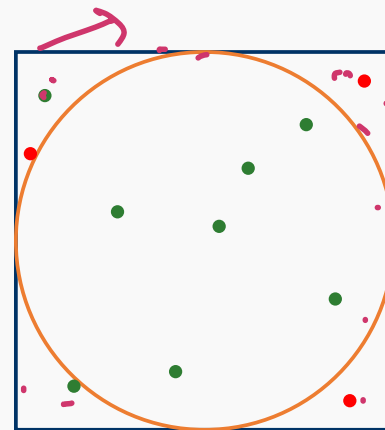
Classic Example: Estimating π by Throwing Darts

- Imagine throwing a dart uniformly at random into the square $[-1, 1] \times [-1, 1]$.



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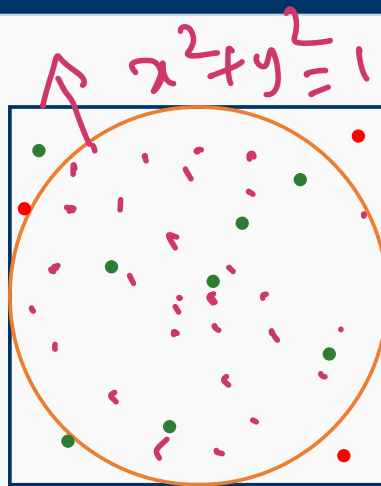
- Imagine throwing a dart uniformly at random into the square $[-1, 1] \times [-1, 1]$.
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- Model the landing point as $(X, Y) \sim \text{Unif}([-1, 1] \times [-1, 1])$. $\Rightarrow$
- Define

$$h(X, Y) = \mathbf{1}\{X^2 + Y^2 \leq 1\} = \begin{cases} 1, & \text{if the dart lands inside the unit circle,} \\ 0, & \text{otherwise.} \end{cases}$$



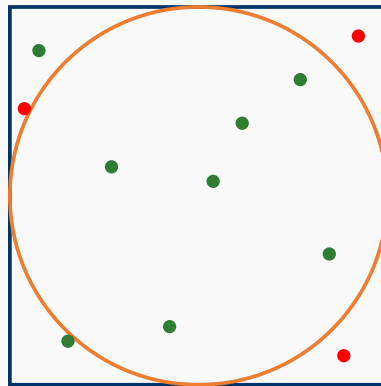
$$\Rightarrow E[h(X, Y)] = \int_{-1}^1 \int_{-1}^1 h(x, y) \frac{1}{4} dx dy$$

$$= 1 \cdot P(X^2 + Y^2 \leq 1) + 0 = P(X^2 + Y^2 \leq 1)$$

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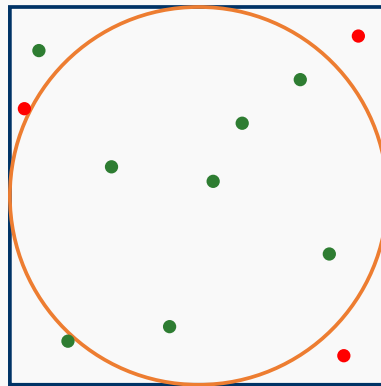
- Since h is an indicator, its expectation is just a probability and under a uniform distribution on the square:

$$E[h(X, Y)] = P(X^2 + Y^2 \leq 1)$$

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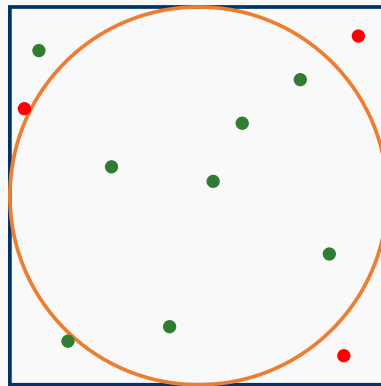
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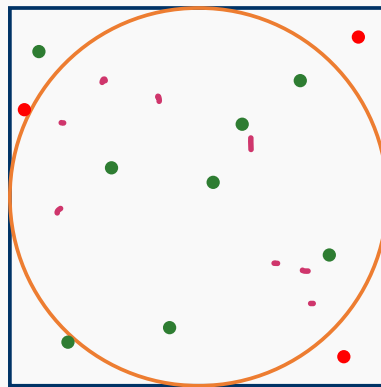
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$$E[h(X, Y)] = P(X^2 + Y^2 \leq 1) = \frac{\text{area inside the circle}}{\text{total area of the square}} = \frac{\pi \cdot 1^2}{(2)(2)} = \left(\frac{\pi}{4}\right)$$

- Therefore $\pi = 4 E[h(X, Y)]$, and the crude MC estimator is

$$\hat{\pi} = 4 \cdot \frac{1}{n} \sum_{i=1}^n h(X_i, Y_i) = 4 \times (\text{fraction of darts inside the circle}).$$

$\xleftarrow{\hspace{1cm}}$ $X_1, Y_1 \sim U[-1, 1]$
 $X_n, Y_n \sim U[-1, 1]$

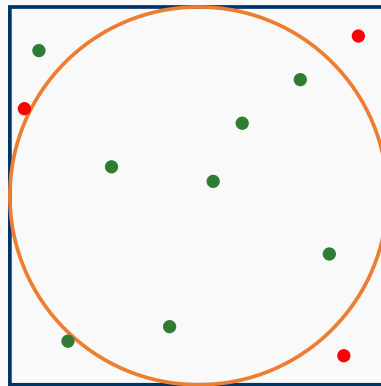
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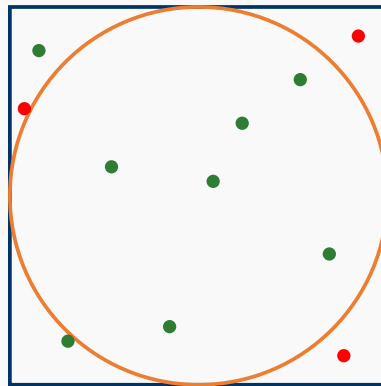
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This is the oldest and most intuitive illustration of the Monte Carlo principle – and it is literally an integral (area) turned into a sample proportion.

Example: An Integral with No Elementary Antiderivative

Target

$$\theta = \int_0^1 e^{-x^2} dx.$$

The integrand e^{-x^2} has **no elementary antiderivative** (it defines the error function, erf) – yet Monte Carlo handles it exactly as easily as any other bounded function.

$$\begin{aligned} \Rightarrow & \Rightarrow \int_0^1 \frac{e^{-x^2}}{1} \cdot 1 dx = \mathbb{E}f[e^{-x^2}] \text{ if } x \sim U[0,1] \\ & \Rightarrow x_1, \dots, x_n \sim U[0,1] \\ \hat{\theta}_n &= \frac{1}{n} \sum_{i=1}^n e^{-x_i^2} \rightarrow \\ & \rightarrow \hat{\theta}_n \text{ estimator of } \theta \end{aligned}$$

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Crude Monte Carlo solution

Write $\theta = \int_0^1 e^{-x^2} \cdot 1 dx = E[h(U)]$ with $U \sim \text{Unif}(0, 1)$ and $h(u) = e^{-u^2}$.

$$\underline{h(u) = e^{-u^2}}$$



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The true value is $\theta \approx 0.7468$; with $n = 10,000$ draws, $\hat{\theta}_n$ typically lands within about ± 0.005 of this. **We will see this in our lab.**

Example: A Genuinely High-Dimensional Integral

Target (engineering / physics flavour)

$$\theta = \int_{[0,1]^{10}} \cos(x_1 + x_2 + \cdots + x_{10}) dx_1 \cdots dx_{10}.$$

A 10-dimensional integral over the unit hypercube – exactly the kind of object that appears in uncertainty quantification for engineering systems with many uncertain inputs.

$$5 \times 5 \times \cdots \times 5 = 5^{10}$$

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$$\Rightarrow \theta = \int_{[0,1]^{10}} \cos(x_1 + \cdots + x_{10}) \, dx_1 \cdots dx_{10}$$

$$= E_f [\cos(x_1 + \cdots + x_{10})]$$

$$\Rightarrow = \frac{1}{n} \sum_{i=1}^n \cos(x_{i1} + \cdots + x_{i10})$$

$$f = \text{Unif}([0,1] \times [0,1] \times \cdots \times [0,1])$$

$$\theta_n^n = \frac{1}{n} \sum_{i=1}^n (\sigma(x_{i1} + \dots + x_{in})) \rightarrow \rightarrow \Theta$$

$$\rightarrow \underset{\sim U[0,1]}{x_{11}}, \underset{\sim U[0,1]}{x_{12}}, \dots, \underset{\sim U[0,1]}{x_{1n}} \sim \dots \sim \underset{\sim U[0,1]}{x} = w[0,1]$$

$$\rightarrow x_{21} \ x_{22} \ x_{2n} \sim \dots \sim$$

$$\rightarrow x_{n1} \ x_{n2} \ x_{nn} \sim \dots \sim$$

$\Rightarrow$ $n \times 10 \approx 10n$ inform

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- Crude Monte Carlo: draw $\mathbf{U}_i = (U_{i1}, \dots, U_{i10}) \stackrel{\text{iid}}{\sim} \text{Unif}[0, 1]^{10}$ and average $\cos(U_{i1} + \cdots + U_{i10})$ – the same $n = 10,000$ draws that worked in 1-D still give a usable answer.

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- This dimension-free behaviour is the reason Monte Carlo dominates high-dimensional integration.

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Law of Large Numbers and Monte Carlo Error ✓

Consistency: The Law of Large Numbers

Quantifying the Error: CLT and Standard Error

Crude Monte Carlo in Practice

Importance Sampling

Variance Reduction Ideas

Practical Monte Carlo Diagnostics

Unit Summary

estimator

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(x_i)$$

to estimator

$$\theta = \int h(x) d\mu$$

$$E(\hat{\theta}_n) = \theta$$

↙ ↘

Why $\hat{\theta}_n$ Actually Works: The LLN

Theorem (Strong Law of Large Numbers) $\Rightarrow$

If $X_1, X_2, \dots \stackrel{\text{iid}}{\sim} f$ and $E_f|h(X)| < \infty$, then

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) \xrightarrow{\text{a.s.}} \theta = E_f[h(X)] \text{ as } n \rightarrow \infty.$$

$\Rightarrow$ almost sure convergence $\Rightarrow$ prob converg

$\Leftarrow$ WL

$\Rightarrow$ Weak Law of Large Number (WLLN)

$\Rightarrow X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F \rightarrow f \rightarrow \theta \rightarrow \text{true mean}$
 $\rightarrow \text{true mean}$

$$\Rightarrow \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{\text{Prob}} \theta, n \rightarrow \infty$$

$$\Rightarrow \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) \xrightarrow{\text{Prob}} \frac{h(\theta)}{h(\theta)} \quad \rightarrow h(\theta^2) < \infty$$

$\Leftarrow$ $\hat{\theta}_n \rightarrow \theta$

SLN $\Rightarrow$ WLLN

Why $\hat{\theta}_n$ Actually Works: The LLN

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- $\rightarrow$ WLLN
- This is what justifies calling $\hat{\theta}_n$ a **consistent** estimator: more simulations $\Rightarrow$ the answer gets arbitrarily close to the truth, with probability 1.

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- ✓ • Unbiasedness (every finite n) and consistency (as $n \rightarrow \infty$) together are the two pillars that make crude Monte Carlo trustworthy.

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- This is what justifies calling $\hat{\theta}_n$ a **consistent** estimator: more simulations $\Rightarrow$ the answer gets arbitrarily close to the truth, with probability 1.
- Unbiasedness (every finite n) and consistency (as $n \rightarrow \infty$) together are the two pillars that make crude Monte Carlo trustworthy.
- The LLN says *nothing* about the finite-sample error, only that it eventually vanishes – for that we need the CLT.

Quantifying Stochastic Error

Suppose we estimate $\theta = E_f[h(X)]$ by the sample mean $\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i)$, where $X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$.
 $\theta = E_f[h(X)] = \int h(x) f(x) dx$

Diagram illustrating the relationship between the integral and the sample mean:

$$\int \frac{h(x)}{f(x)} f(x) dx \quad \leftarrow \hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i) \quad \leftarrow X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f$$

Handwritten definition of $h(x)$:

$$h(x) = \frac{g(x)}{f(x)}$$

CLT $\Rightarrow X_1, \dots, X_n \stackrel{\text{iid}}{\sim} F, \mu, \sigma^2 < \infty$
 (where μ is the mean and σ^2 is the variance)

$\Rightarrow \frac{1}{n} \sum_{i=1}^n X_i = \bar{X}_n \Rightarrow \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \sim N(0,1)$

$\Rightarrow \bar{X}_n \sim N[\mu, \sigma^2/n]$

Quantifying Stochastic Error

Suppose we estimate $\theta = E_f[h(X)]$ by the sample mean

$$\hat{\theta}_n = \frac{1}{n} \sum_{i=1}^n h(X_i), \quad X_1, \dots, X_n \stackrel{\text{iid}}{\sim} f.$$

- **Central Limit Theorem:** if $\sigma^2 = \text{Var}_f[h(X)] < \infty$, then for large n ,

$$\hat{\theta}_n \stackrel{\text{approx}}{\sim} N\left(\theta, \frac{\sigma^2}{n}\right)$$

- **Monte Carlo Standard Error:**

$$\text{SE}(\hat{\theta}_n) = \frac{\sigma}{\sqrt{n}} \approx \frac{s}{\sqrt{n}},$$

where s is the sample standard deviation of $h(X_1), \dots, h(X_n)$.

$$\Rightarrow \hat{\theta}_n \pm Z_{\alpha/2} \text{SE}(\hat{\theta}_n)$$

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Approximate $(1 - \alpha)$ confidence interval

$$\hat{\theta}_n \pm z_{\alpha/2} \cdot \frac{s}{\sqrt{n}}.$$